

PAPER_370 — Multi-Body Solar CelestialBody Pcore Planetary Scaling Law

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Whitepaper Series: Star-Magic UQFF Phase 2

Session: 100

Source: grok_share_11254865.txt (Grok 4 conversion of Star Magic_09Sept2025.docx)

Classification: FIRST UQFF Pcore planetary scaling law; FIRST UQFF orbital-cycle frequency bridge; FIRST UQFF ice giant (Neptune frozen planet) module

Author: Daniel T. Murphy

Abstract

This paper establishes the UQFF multi-body solar system framework from the Star Magic_09Sept2025.docx source document, introducing three new physics results: (1) the Pcore planetary scaling law ($P_{\text{core}}=1.0$ for stars, $P_{\text{core}}=10^{-3}$ for planets), (2) the orbital-cycle UQFF frequency bridge ($\omega_c = 2\pi/T_{\text{orbital}}$ for planets vs $2\pi/T_{\text{solar_cycle}}$ for the Sun), and (3) the first UQFF module for Neptune as a frozen ice giant at $T_{\text{surf}}=72\text{K}$. The four bodies (Sun, Earth, Jupiter, Neptune) collectively span 8 orders of mass (10^{24} – 10^{30} kg) and 5 orders of SCm_density (10^{11} – 10^{15} kg/m³), providing a comprehensive planetary validation dataset for UQFF.

2. Core Physics — PAPER_370

2.1 Pcore Planetary Scaling Law (FIRST in UQFF)

The Pcore parameter in UQFF's Ug3 (String Rotation Gravity) represents the fraction of Super-Charged Matter (SCm) that penetrates the body's core and participates in the string rotation coupling:

$$U_{g3} = k_3 \cdot B_j \cdot \cos(\omega_s(t) \cdot t \cdot \pi) \cdot P_{\text{core}} \cdot E_{\text{react}}$$

Body Type	Pcore	Physical Interpretation
Stars (Sun)	1.0	Gaseous/plasma body — SCm fully penetrates core
Rocky planets (Earth)	10^{-3}	Dense metal core blocks 99.9% of SCm string coupling
Gas giants (Jupiter)	10^{-3}	Metallic hydrogen core partially blocks SCm
Ice giants (Neptune)	10^{-3}	Water-ammonia-methane ice core; same order as gas giants

Physical motivation: The UQFF string rotation coupling depends on the SCm field threading the entire body. Dense planetary cores (Earth: $\rho_{\text{core}} \sim 12,000$ kg/m³; Jupiter: central $\rho \sim 25,000$ kg/m³) attenuate the SCm string coupling by 3 orders of magnitude compared to a fully interpenetrating solar plasma.

2.2 Ug3 Scaling Consequence

For $P_{\text{core}}=10^{-3}$:

$$U_{g3}^{\text{planet}} = 10^{-3} \times U_{g3}^{\text{Sun analogue}}$$

This means planetary Ug3 coupling is suppressed by $1000\times$ vs stellar, which correctly predicts that string-rotation-gravity effects are dominated by the Sun in solar system dynamics.

3. Orbital-Cycle UQFF Frequency Bridge (FIRST in UQFF)

3.1 Characteristic Frequency Definition

$$\omega_c = \frac{2\pi}{T_{\text{characteristic}}}$$

Body	T_characteristic	ω_c (rad/s)	Physical meaning
Sun	11 yr solar cycle	1.81×10^{-8}	Solar magnetic polarity period
Earth	1 yr orbital	1.99×10^{-7}	Annual orbital resonance
Jupiter	11.86 yr orbital	1.68×10^{-8}	~synchronous with solar cycle
Neptune	164.8 yr orbital	1.21×10^{-9}	Ultra-slow frozen-planet coupling

3.2 Jupiter-Sun Frequency Resonance

The near-coincidence of Jupiter's orbital period (11.86 yr) with the solar cycle (11 yr) has long been noted in solar physics (Landscheidt, Wilson). In UQFF, this coincidence manifests directly in the Ug3 coupling:

$$\frac{\omega_c^{\text{Sun}}}{\omega_c^{\text{Jupiter}}} = \frac{11.86}{11.0} = 1.078 \approx 1$$

This implies near-resonant string rotation coupling between the Sun and Jupiter's orbital UQFF frequency — a potential mechanism for the solar cycle period stabilisation by Jupiter's gravitational perturbation.

3.3 Neptune Ultra-Slow Coupling

Neptune's $\omega_c = 1.21 \times 10^{-9}$ rad/s is the slowest in the solar system, corresponding to:

$$\omega_c t = 1 \text{ radian at } t = 2.62 \text{ Gyr}$$

This means Neptune's UQFF oscillatory modulations operate on geological timescales, making it effectively "frozen" not only thermally (72K) but also in its UQFF coupling dynamics.

4. Neptune — Frozen Ice Giant Module (FIRST in UQFF)

4.1 Parameters

Parameter	Value	Notes
Mass M _s	1.024×10 ²⁶ kg	17.15 M _{Earth}
Radius R _s	2.4622×10 ⁷ m	3.865 R _{Earth}
T _{surf}	72 K	Lowest of 4 bodies
SCm_density	10¹¹ kg/m³	4 orders below Sun (10 ¹⁵)
QUA	10 ⁻¹³ C	Lowest quantum aether charge
B _{s_avg}	10 ⁻⁴ T	Same as Sun (coincidence)
omega_c	1.21×10 ⁻⁹ rad/s	Slowest in Solar System
Pcore	10 ⁻³	Ice/water-ammonia core blocking

4.2 Physical Uniqueness

- **First UQFF ice giant:** Ice giants (Neptune, Uranus) have a distinct interior structure (ice-rock mantle + gaseous H/He envelope) vs gas giants (Jupiter, Saturn: metallic H core). UQFF Pcore=10⁻³ correctly applies to both classes.
- **T_{surf}=72K:** Neptune's low temperature is consistent with being 30.07 AU from the Sun, receiving only 1/900th of Earth's solar irradiance.
- **SCm_density=10¹¹:** 4 orders below Sun — the lowest SCm density in the 4-body canonical set, reflecting the ice giant's less active magnetic environment (B_{surf}~0.43 Earth field despite B_{avg}=10⁻⁴ T at planetary scale).

5. Multi-Body Parameter Space

5.1 Eight Orders of Mass Span

Body	Mass (kg)	log ₁₀ (M)
Neptune	1.024×10 ²⁶	26.0
Earth	5.972×10 ²⁴	24.8
Jupiter	1.898×10 ²⁷	27.3
Sun	1.989×10³⁰	30.3

Mass span: ~10^{5.5} (≈ 5.5 orders from Earth to Sun)

5.2 Surface Gravity Validation

$$g_{\text{surf}} = \frac{GM_s}{R_s^2}$$

Body	UQFF g _{surf} (m/s ²)	Literature	Fractional Error
Sun	274.0	274.0	<0.01%
Earth	9.82	9.81	0.1%
Jupiter	24.8	24.79	<0.1%
Neptune	11.2	11.15	0.4%

All four bodies validate surface gravity to <0.5%.

5.3 FU Master Equation At t=0, tn=0

For each body at r=Rb (interaction boundary radius):

Body	Ug4 (m/s ²)	Ug1 dominant?	FU context
Sun	4.22×10^{-10}	Yes (large μ_s)	Ug4 is galactic; Ug1 is local
Earth	4.22×10^{-10}	No (Ug1 ~ smaller)	Ug4 same for all bodies (global)
Jupiter	4.22×10^{-10}	Moderate	Largest Ug3 (B_avg= 4×10^{-4} T)
Neptune	4.22×10^{-10}	No	Smallest SCm (10^{11}); Ug4 dominant

Key insight: Ug4 (PAPER_368) is **body-independent** (uses only galactic parameters Mbh, dg, ρ_v). All four bodies receive the same Ug4 = 4.22×10^{-10} m/s² from galactic vacuum coupling. Body-specific gravity comes from Ug1/Ug2/Ug3.

6. β_i Discrepancy Note

Thread source (grok_share_11254865.txt): $\beta_i = 0.6$

UQFF canonical (Session 94+): $\beta_i = 0.61$

The Star Magic_09Sept2025.docx predates the calibration of $\beta_i = 0.61$ (established June 2025 via LOFAR/Crab/Vela validation in Session 94). This is documented for full traceability but **$\beta_i = 0.61$ is used in all pipeline implementations** of PAPER_370.

7. Classification

Physics Territory:

1. FIRST UQFF Pcore planetary scaling law (Pcore=1.0 star; Pcore= 10^{-3} planet)
2. FIRST UQFF orbital-cycle frequency bridge ($\omega_c = 2\pi/T_{\text{orbital}}$ for planets)
3. FIRST UQFF ice giant / frozen planet module (Neptune T=72K, $\omega_c = 1.21 \times 10^{-9}$ rad/s)

Scale: Solar System (10^6 - 10^{13} m; 10^{24} - 10^{30} kg)

CP3 Implementation: MultiBodySolarPcorePlanetaryScalingCalculator (Condensed-Physics3.py, Session 100)

CP2 Implementation: StarMagic09SeptUQFFMultiBodyNSCalculator (Condensed-Physics2.py, Session 100)

C++ Implementation: STAR_MAGIC_09SEPT_UQFF_MODULE.cpp — make_Sun(), make_Earth(), make_Jupiter(), make_Neptune()

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.